WORKED SOLUTION

STEP III 2003, Question 3

Pure • Curve Sketching

STEP III 2003, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/03-s3-q3

The key idea

Write $u = (1+x)^m$ and $v = (1-x)^m$. Then $\mathrm{f} = (u-v)/(u+v)$, and the quotient rule collapses beautifully because $u'v - uv'$ has a common factor $(1-x^2)^{m-1}$. That derivative is never negative, which tells us almost everything about the shape of $y = \mathrm{f}(x)$. For the reciprocal curve there is a hidden symmetry: replacing x by $1/x$ turns f into $1/\mathrm{f}$.

Solution

The denominator is never zero

Suppose $(1+x)^m + (1-x)^m = 0$ for some real x.

If m is even, both terms are non-negative, so both must be zero. That needs $x = -1$ and $x = 1$ at the same time, which is impossible.

If m is odd, then $(1+x)^m = -(1-x)^m = (x-1)^m$. The map $t \mapsto t^m$ is strictly increasing on $\mathbb{R}$ when m is odd, so it is one-to-one, and $1+x = x-1$, that is, $1 = -1$. This is a contradiction.

So $(1+x)^m + (1-x)^m \neq 0$ for every real x, and f is defined and differentiable on all of $\mathbb{R}$.

The derivative

With $u = (1+x)^m$ and $v = (1-x)^m$ we have $u' = m(1+x)^{m-1}$ and $v' = -m(1-x)^{m-1}$. By the quotient rule,

$$\mathrm{f}'(x) = \frac{(u'-v')(u+v) - (u-v)(u'+v')}{(u+v)^2} = \frac{2(u'v - uv')}{(u+v)^2}.$$

The numerator simplifies:

$$\begin{aligned} u'v - uv' &= m(1+x)^{m-1}(1-x)^m + m(1+x)^m(1-x)^{m-1} \\ &= m(1+x)^{m-1}(1-x)^{m-1}[(1-x)+(1+x)] \\ &= 2m(1-x^2)^{m-1}. \end{aligned}$$

Therefore

$$\mathrm{f}'(x) = \frac{4m\,(1-x^2)^{m-1}}{[(1+x)^m + (1-x)^m]^2}.$$

The case $m = 5$: the curve $y = \mathrm{f}(x)$

Expanding with the binomial theorem, the even powers cancel in the numerator and the odd powers cancel in the denominator:

$$\mathrm{f}(x) = \frac{2(5x + 10x^3 + x^5)}{2(1 + 10x^2 + 5x^4)} = \frac{x(x^4 + 10x^2 + 5)}{5x^4 + 10x^2 + 1}, \qquad \mathrm{f}'(x) = \frac{20(1-x^2)^4}{[(1+x)^5 + (1-x)^5]^2}.$$

Now we read off the features.

- *Symmetry.* $\mathrm{f}(-x) = -\mathrm{f}(x)$, so the curve has rotational symmetry about the origin. It passes through $(0, 0)$ with gradient $\mathrm{f}'(0) = 20/2^2 = 5$.
- *Monotonic.* $\mathrm{f}'(x) \geq 0$ for all x, with equality only at $x = \pm 1$. So f is increasing everywhere. At $x = \pm 1$ it has stationary points, and since $(1-x^2)^4$ does not change sign these are points of inflection, not turning points. They are at $(1, 1)$ and $(-1, -1)$: the $(1-x)^5$ or $(1+x)^5$ term vanishes, leaving $\mathrm{f} = \pm 1$.
- *Large $|x|$.* $\mathrm{f}(x) \approx x/5$. More precisely,

  $$\mathrm{f}(x) - \frac{x}{5} = \frac{x(40x^2 + 24)}{5(5x^4 + 10x^2 + 1)},$$

  which tends to 0 as $|x| \to \infty$. So $y = x/5$ is an oblique asymptote. The curve lies above it for $x > 0$ and below it for $x < 0$.

Ask yourself Where is $\mathrm{f}(x)$ bigger than 1 in size, and where is it smaller?

Since f is increasing with $\mathrm{f}(\pm 1) = \pm 1$, we have $|\mathrm{f}(x)| < 1$ for $|x| < 1$ and $|\mathrm{f}(x)| > 1$ for $|x| > 1$. This matters for the reciprocal. (For $|x| < 1$, putting $x = \tanh t$ gives $(1+x)/(1-x) = \mathrm{e}^{2t}$, and dividing top and bottom of f by $(1-x)^5$ shows $\mathrm{f}(\tanh t) = \tanh 5t$. That is another way to see the S-shape between -1 and 1.)

The curve $y = 1/\mathrm{f}(x)$

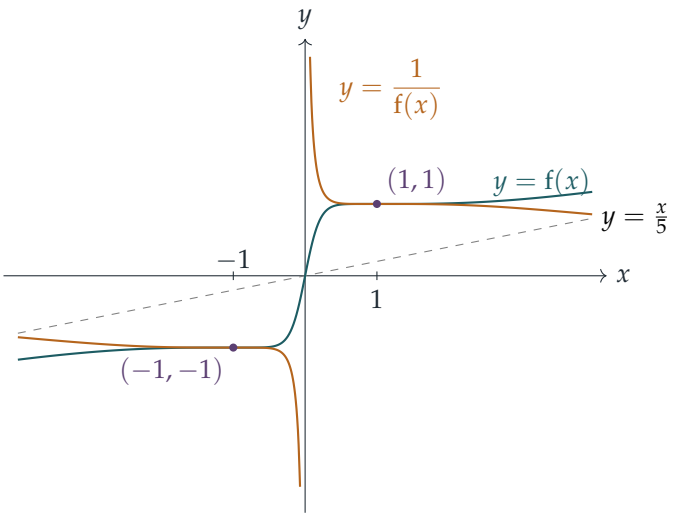
Replacing x by $1/x$ and multiplying top and bottom by x^5 gives

$$\mathrm{f}(1/x) = \frac{(x+1)^5 - (x-1)^5}{(x+1)^5 + (x-1)^5} = \frac{(1+x)^5 + (1-x)^5}{(1+x)^5 - (1-x)^5} = \frac{1}{\mathrm{f}(x)},$$

using $(x-1)^5 = -(1-x)^5$. So $y = 1/\mathrm{f}(x)$ is the curve $y = \mathrm{f}(1/x)$, and its features follow from those of f:

- It is odd, and undefined at $x = 0$, the only zero of f. Near 0, $1/\mathrm{f}(x) \approx 1/(5x)$, so the y-axis is a vertical asymptote, with $y \to +\infty$ as $x \to 0^+$ and $y \to -\infty$ as $x \to 0^-$.
- Its derivative is $-\mathrm{f}'(x)/\mathrm{f}(x)^2 \le 0$, so it is decreasing on $x > 0$ and on $x < 0$. It has stationary points of inflection at $(1,1)$ and $(-1,-1)$.
- As $|x| \to \infty$, $1/\mathrm{f}(x) \approx 5/x \to 0$, so the x-axis is an asymptote. The curve lies above it for $x > 0$ and below it for $x < 0$.

The two curves meet only where $\mathrm{f}(x)^2 = 1$, that is, at $(1,1)$ and $(-1,-1)$. Both curves are horizontal there, so they touch rather than cross.



Checks

- Take $m = 1$: the formula gives $\mathrm{f}'(x) = 4/2^2 = 1$, and indeed $\mathrm{f}(x) = 2x/2 = x$.
- Numerically, $\mathrm{f}(1000)/1000 \approx 0.2000016$, consistent with the asymptote $y = x/5$. A central-difference derivative agrees with the formula for f′ at random points for $m = 1, \ldots, 8$.
- $\mathrm{f}(\tanh t) = \tanh 5t$ and $\mathrm{f}(1/x) = 1/\mathrm{f}(x)$ both check out to 10^{-9} at random points.

Common slips

- Calling $x = \pm 1$ maxima or minima. The derivative touches zero there but does not change sign, so these are stationary points of inflection.
- Missing the oblique asymptote: f is a quintic over a quartic, so it grows like $x/5$ rather than levelling off.

- Drawing $y = 1/\mathrm{f}(x)$ crossing $y = \mathrm{f}(x)$ at $(\pm 1, \pm 1)$. Both curves are flat there, so they touch.

Beyond STEP

The substitution $x = \tanh t$ shows that f is the formula for $\tanh 5t$ in terms of $\tanh t$, the hyperbolic version of the multiple-angle formula for $\tan 5\theta$. Seen this way, f is a rational map that conjugates to multiplication by 5. Maps like this, and how they behave under iteration, are a starting point of complex dynamics. Needham's *Visual Complex Analysis* is a beautiful place to see how such rational functions act on the whole complex plane.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.